Supplementary Information

A Snapshot of the Repeated Trust Game as Seen by Participants

Figure S1 shows a screenshot of the repeated Trust Game at the moment the participant is required to make a decision of how much to send back to the Investor.

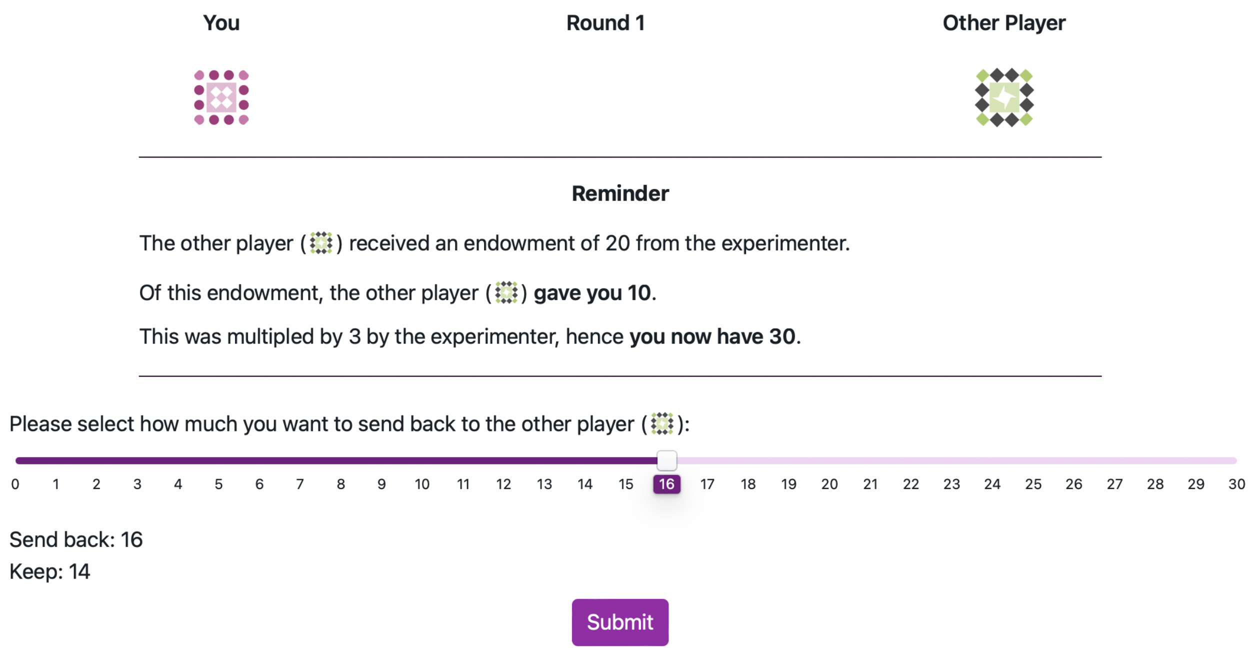

Figure S1: Screenshot of the RTG as seen by participants at the decision phase.

Hidden Markov Model Used to Simulate the Investor's Actions

The HMM assumes that the probability of each investment $I_t = 0, \ldots, 20$, at each trial t, conditional on the current state of the investor S_t, is dependent on an underlying normal distribution with mean μ_s and standard deviation σ_s. The probability of each discrete investment was determined from the cumulative normal distribution Φ, computing the probability of a Normal variate falling between the midway points of the response options. As responses were bounded at 0 and 20, we normalized these probabilities further by taking the endpoints into account. For instance, the probability of an investment $I_t = 2$ is defined as:

$$P(I_t = 2 | S_t = s) = \frac{\Phi(2.5|\mu_s, \sigma_s) - \Phi(1.5|\mu_s, \sigma_s)}{\Phi(20.5|\mu_s, \sigma_s) - \Phi(-0.5|\mu_s, \sigma_s)}$$

Note that the denominator truncates the distribution between 0 and 20. To estimate the transition probability between states for the investor, a multinomial logistic regression model was fitted to the investor's data such as:

$$P(S_{t+1} = s'|S_t = s, X_t = x) = \frac{\exp(\beta_{0,s,s'} + \beta_{1,s,s'}x)}{\sum_{s''} \exp(\beta_{0,s,s''} + \beta_{1,s,s''}x)}$$

where $X_t = R_t - I_t$ is the net return to the investor with R_t the amount returned by the trustee and I_t is the Investment sent.

The advantages of this approach is that it does not require any a priori assumptions about the model features. The number of states, the policy conditional on the state, and the transition function between states can all determined in a purely data-driven way. These HMMs can in turn be used to simulate a human-like agent playing the trust game. This agent may transition to a new state depending on the other player's actions and adopt a policy reflecting its state, thus simulating changes in emotional dispositions of human players during a repeated game. When the investor gains from the interaction, they become more likely to transition to a state where their policy is more "trusting" with generally higher investments. However, faced with losses, the investor is more likely to transition to a more cautious policy with generally lower investments. The policies and the transitions between states are sufficient to build an agent that reflects this type of adaptive behavior and reacts to the trustee's action choices in a way that mimics a human player.

We estimated a three-state model for investor's behaviour, using maximum likelihood estimation via the Expectation-Maximisation algorithm as implemented in the depmixS4 package for R (Visser and Speekenbrink 2021). The model was estimated using investments from existing datasets of human dyads playing 10 rounds of the RTG with the same trustee. The dataset consisted of a total of 381 games from two data sources: First, a total of 93 repeated trust games with healthy investors and a mix of healthy trustees and trustees diagnosed with Borderline Personality Disorder (BPD) (King-Casas et al. 2008). The second source was from data collected as part of a project investigating social exchanges in BPD and antisocial personality disorder reported on elsewhere (Euler et al. 2021; Huang et al. 2020; Rifkin-Zybutz et al. 2021) and consists of 288 games. In both datasets, the investor on which we modelled the HMM's strategy was always selected from a healthy population and the trustees were a mix of healthy participants and those with personality disorders allowing for a diversified interaction behavior.

Mixed-effects Models for Participant Returns

We fit a linear mixed-effects model to participant returns as a proportion of the multiplied investment received. The model specification is described below, and the results are presented in Table 1.

$$\begin{aligned}
\text{ReturnPercentage}_{ij} = &\beta_0 + \beta_1\text{Opponent}_i + \beta_2\text{Investment}_i + \beta_3\text{D-level}_i + \beta_4\text{Order}_i + \beta_5\text{Round}_i + \\
&[\text{All interaction terms}] + \\
&b_{0j} + b_{1j}\text{Opponent}_i + b_{2j}\text{Investment}_i + \epsilon_{ij}
\end{aligned}$$

where:

- $\text{ReturnPercentage}_{ij}$: The percentage of the tripled investment returned by participant j on trial i.
- The **Fixed Effects** are:
  - β_0: The overall intercept.
  - β_1 to β_5: The main effects for Opponent Type (Opponent_i), scaled Investment (Investment_i), D-Factor level (D-level_i), presentation order (Order_i), and round number (Round_i).
  - The model includes all possible two-way, three-way, four-way, and five-way interaction terms among the five fixed effects, represented by [All interaction terms].
- The **Random Effects** account for by-participant variability:
  - b_{0j}: A random intercept for each participant j.
  - b_{1j}: A random slope for the effect of Opponent Type for each participant j.
  - b_{2j}: A random slope for the effect of Investment for each participant j.
- ϵ_{ij}: The residual error term for participant j on trial i.

Table 1: Full Results of the Linear Mixed-Effects Model of Participant Percentage Returns.

Term	Estimate	Std. Error	df	t-value	p-value	
(Intercept)	0.460	0.015	199.901	30.771	0.000	***
Opponent	-0.001	0.006	669.512	-0.135	0.892	
Investment	0.021	0.009	341.738	2.341	0.020	*
D-Level	-0.010	0.015	199.901	-0.653	0.514	
Order	0.000	0.021	199.889	0.006	0.996	
Round	-0.002	0.000	8140.419	-4.733	0.000	***
Opponent x Investment	-0.002	0.006	8173.610	-0.363	0.716	
Opponent x D-Level	-0.004	0.006	669.512	-0.672	0.502	
Investment x D-Level	0.002	0.009	341.738	0.169	0.866	
Opponent x Order	0.003	0.008	667.589	0.370	0.712	
Investment x Order	-0.002	0.013	344.063	-0.143	0.886	
D-Level x Order	0.013	0.021	199.889	0.648	0.518	
Opponent x Round	0.000	0.000	8154.545	-0.663	0.507	
Investment x Round	0.000	0.000	8189.035	-1.082	0.279	
D-Level x Round	0.000	0.000	8140.419	-1.027	0.304	
Order x Round	0.001	0.000	8135.490	2.234	0.026	*
Opponent x Investment x D-Level	0.012	0.006	8173.610	2.182	0.029	*
Opponent x Investment x Order	0.007	0.008	8134.185	0.870	0.384	
Opponent x D-Level x Order	-0.005	0.008	667.589	-0.663	0.508	
Investment x D-Level x Order	-0.021	0.013	344.063	-1.650	0.100	
Opponent x Investment x Round	0.000	0.000	8207.016	0.402	0.688	
Opponent x D-Level x Round	0.000	0.000	8154.545	0.548	0.584	
Investment x D-Level x Round	-0.001	0.000	8189.035	-2.766	0.006	**
Opponent x Order x Round	0.000	0.000	8144.127	0.553	0.580	
Investment x Order x Round	0.000	0.001	8201.183	0.280	0.779	
D-Level x Order x Round	-0.001	0.000	8135.490	-1.140	0.254	
Opponent x Investment x D-Level x Order	-0.002	0.008	8134.185	-0.273	0.785	
Opponent x Investment x D-Level x Round	-0.001	0.000	8207.016	-1.638	0.102	
Opponent x Investment x Order x Round	-0.001	0.001	8206.907	-1.651	0.099	
Opponent x D-Level x Order x Round	0.000	0.000	8144.127	0.654	0.513	
Investment x D-Level x Order x Round	0.002	0.001	8201.183	3.753	0.000	***
Opponent x Investment x D-Level x Order x Round	0.000	0.001	8206.907	-0.069	0.945	

Significance codes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Mixed Effects Models for Opponent Ratings

We fit linear mixed-effects models to participants' ratings of their opponents (cooperativeness, willingness to play again, and trustworthiness). The model specification for the cooperative ratings is described below as an example; identical model structures were used for the other ratings.

$$\begin{aligned}\text{CooperativeRating}_{ij} = &\beta_0 + \beta_1\text{D-level}_i + \beta_2\text{Order}_i + \beta_3\text{Opponent}_i +\\ &[\text{Interaction terms}] +\\ &b_{0j} + \epsilon_{ij}\end{aligned}$$

where:

- $\text{CooperativeRating}_{ij}$: The cooperativeness rating given by participant j for opponent i.
- The **Fixed Effects** are:
  - β_0: The overall intercept.

- β_1 to β_3: The main effects for D-Factor level (D-level$_i$), game order (Order$_i$), and Opponent Type (Opponent$_i$).
- The model includes all two-way and three-way interaction terms among the three fixed effects, represented by [Interaction terms].

- The **Random Effects** account for by-participant variability:
  - b_{0j}: A random intercept for each participant j.
- ϵ_{ij}: The residual error term for participant j's rating of opponent i.

Detailed Results

Cooperative Ratings

Linear mixed-effects analysis revealed a significant main effect of D-level, $F(1, 179) = 7.35$, $p = .007$, with Low-D participants rating their opponents as more cooperative than High-D participants. A significant main effect of game order, $F(1, 179) = 10.67$, $p = .001$, indicated participants rated opponents in their first game as more cooperative than those in their second game. Additionally, there was a significant main effect of opponent type, $F(1, 179) = 5.31$, $p = .022$, with Human-like HMM opponents receiving higher cooperative ratings than Responsive opponents. No significant interactions were observed (all ps $> .29$).

Play Again Ratings

Analysis of participants' willingness to play with the same opponent again revealed a significant main effect of game order, $F(1, 179) = 13.83$, $p < .001$, with participants generally more willing to play again with opponents from their first game. This main effect was qualified by a significant D-level $\times$ Game Order interaction, $F(1, 179) = 4.05$, $p = .046$.

Post-hoc analyses revealed that in the first game, High-D participants were significantly less willing to play again with their opponents compared to Low-D participants ($\Delta M = -1.07$, 95% CI $[-1.96, -0.18]$, $t(320.47) = -2.37$, $p = .018$), while no such difference existed in the second game ($\Delta M = -0.03$, 95% CI $[-0.92, 0.86]$, $t(320.47) = -0.07$, $p = .947$). Examining changes across games, Low-D participants showed a significant decrease in willingness to play again from the first to the second game ($\Delta M = 1.48$, 95% CI $[0.76, 2.21]$, $t(179) = 4.05$, $p < .001$), while High-D participants maintained consistent ratings across games ($\Delta M = 0.44$, 95% CI $[-0.28, 1.16]$, $t(179) = 1.21$, $p = .229$).

Trusting Ratings

For trust ratings, significant main effects were observed for D-level, $F(1, 179) = 7.53$, $p = .007$; game order, $F(1, 179) = 7.21$, $p = .008$; and opponent type, $F(1, 179) = 4.62$, $p = .033$. These effects were qualified by a significant three-way interaction between D-level, game order, and opponent type, $F(1, 179) = 4.76$, $p = .030$.

Post-hoc analyses revealed a complex pattern of trust perceptions. High-D participants rated Human-like opponents as significantly less trusting than Low-D participants in the first game ($\Delta M = -1.65$, 95% CI $[-2.75, -0.55]$, $t(314.81) = -2.96$, $p = .003$). In contrast, High-D participants rated Responsive opponents as significantly less trusting than Low-D participants in the second game ($\Delta M = -1.51$, 95% CI $[-2.61, -0.41]$, $t(314.81) = -2.70$, $p = .007$). Low-D participants showed a significant *decrease* in trust ratings for Human-like opponents from the first to the second game ($\Delta M = 1.22$, 95% CI $[0.14, 2.29]$, $t(314.81) = 2.22$, $p = .027$), while High-D participants similarly showed a significant *decrease* in trust ratings for Responsive opponents from the first to the second game ($\Delta M = 1.35$, 95% CI $[0.27, 2.43]$, $t(314.81) = 2.47$, $p = .014$). Additionally, High-D participants in the second game differentiated between opponent types, rating Human-like opponents as significantly more trusting than Responsive opponents ($\Delta M = 1.37$, 95% CI $[0.30, 2.45]$, $t(314.81) = 2.51$, $p = .013$).

Summary

Participants with higher Dark Factor scores demonstrated consistently more negative perceptions of their opponents, particularly regarding cooperation and trust. The pattern of results indicates that individual differences in Dark Factor traits influence not only the overall level of opponent ratings but also how these ratings change across repeated interactions and between different opponent types. Notably, Low-D participants showed greater sensitivity to game order, with more pronounced decreases in ratings from first to second game, while High-D participants demonstrated greater discrimination between opponent types in their trust ratings.

Sensitivity Analysis: Excluding Participants Who Believed They Played a Computer

To address concerns about the ecological validity of our findings given that some participants suspected they were playing against a computer rather than a human, we conducted a sensitivity analysis restricting the sample to participants who either believed they were playing against a human or were unsure. This approach excludes only those participants who were certain they played against a computer (N = 79, 43.2%), retaining participants who believed "Human" (N = 47, 25.7%) or indicated "Not Sure" (N = 57, 31.1%), for a total of 104 participants (56.8% of the original sample).

We re-estimated our primary mixed-effects model on this restricted sample using the same specification as the full analysis.

Table 2: Sensitivity Analysis: Mixed-Effects Model Results for Participants Who Believed Human or Were Unsure (N = 104).

Term	Estimate	Std. Error	df	t-value	p-value	
(Intercept)	0.459	0.021	113.413	22.276	0.000	***
Opponent	-0.006	0.009	345.156	-0.665	0.506	
Investment	0.023	0.014	184.229	1.607	0.110	
D-Level	-0.030	0.021	113.413	-1.435	0.154	
Order	-0.007	0.027	113.526	-0.242	0.809	
Round	-0.002	0.000	4595.593	-3.964	0.000	***
Opponent x Investment	-0.002	0.009	4659.198	-0.192	0.848	
Opponent x D-Level	0.008	0.009	345.156	0.922	0.357	
Investment x D-Level	0.000	0.014	184.229	0.012	0.991	
Opponent x Order	0.006	0.012	343.164	0.532	0.595	
Investment x Order	-0.006	0.019	184.927	-0.318	0.751	
D-Level x Order	0.040	0.027	113.526	1.470	0.144	
Opponent x Round	0.000	0.000	4608.679	-0.049	0.961	
Investment x Round	-0.001	0.001	4635.621	-1.506	0.132	
D-Level x Round	0.000	0.000	4595.593	-0.431	0.666	
Order x Round	0.001	0.001	4597.018	2.027	0.043	*
Opponent x Investment x D-Level	0.023	0.009	4659.198	2.676	0.007	**
Opponent x Investment x Order	0.003	0.011	4635.260	0.240	0.810	
Opponent x D-Level x Order	-0.026	0.012	343.164	-2.275	0.024	*
Investment x D-Level x Order	-0.009	0.019	184.927	-0.487	0.627	
Opponent x Investment x Round	0.000	0.001	4655.151	-0.107	0.915	
Opponent x D-Level x Round	-0.001	0.000	4608.679	-1.155	0.248	
Investment x D-Level x Round	-0.001	0.001	4635.621	-2.314	0.021	*
Opponent x Order x Round	0.000	0.001	4605.357	0.117	0.907	
Investment x Order x Round	0.001	0.001	4638.753	1.804	0.071	
D-Level x Order x Round	-0.001	0.001	4597.018	-1.500	0.134	
Opponent x Investment x D-Level x Order	-0.010	0.011	4635.260	-0.899	0.369	
Opponent x Investment x D-Level x Round	-0.001	0.001	4655.151	-1.345	0.179	
Opponent x Investment x Order x Round	0.000	0.001	4648.569	-0.443	0.658	
Opponent x D-Level x Order x Round	0.002	0.001	4605.357	2.474	0.013	*
Investment x D-Level x Order x Round	0.002	0.001	4638.753	2.831	0.005	**
Opponent x Investment x D-Level x Order x Round	0.000	0.001	4648.569	0.012	0.991	

Significance codes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The sensitivity analysis reveals that the key patterns observed in the full sample remain consistent in this restricted sample. Specifically, the D-Level × Opponent × Investment interaction—our primary finding demonstrating that high-D individuals show differential exploitation patterns depending on opponent type—

shows the same direction of effect (positive estimate: $\beta = 0.023$, $t = 2.676$, $p = 0.007$). While reduced statistical power from the smaller sample affects significance levels for some effects, the overall pattern of results supports our conclusions that the observed D-factor effects are not artifacts of participants' beliefs about their opponents.

Growth Model Comparison: Linear vs. Polynomial Trends

Following reviewer suggestions, we evaluated whether adding a quadratic trend for round number would improve model fit and reveal differential learning trajectories between D-level groups. We compared our primary linear model with a growth model that included both linear and quadratic round terms, with the quadratic term interacting with D-level and opponent type.

Model Comparison Results

Likelihood ratio test:

We used a likelihood ratio test to formally compare the linear and growth models. Models were refit using maximum likelihood estimation for valid comparison.

The likelihood ratio test indicated that the growth model (with quadratic terms) did significantly improve fit over the linear model ($\chi^2(4) = 15.05$, $p = 0.005$).

Model fit indices:

Model	AIC	BIC
Linear (primary)	-8049.7	-7774.7
Growth (quadratic)	-7982.1	-7678.9

Note: Lower values indicate better fit. The BIC, which penalizes model complexity more heavily than AIC, favors the linear model by 96 points.

Key findings from the growth model:

1. **Quadratic term:** The quadratic trend was statistically significant ($F = 9.06$, $p = 0.003$), indicating some curvature in the overall trajectory of return percentages across rounds.
2. **D-level × Quadratic interaction:** Critically, D-level did *not* interact with the quadratic term ($F = 0.46$, $p = 0.497$). This indicates that high-D and low-D participants showed similar curvature in their trajectories—they differ in their linear slopes (rate of change), not in the shape of change over time.
3. **Core three-way interaction:** The D-level × Opponent × Investment interaction became marginally significant in the growth model ($F = 3.81$, $p = 0.051$), compared to $p = .004$ in the linear model.

Conclusion: Although the likelihood ratio test reached significance, the absence of a theoretically meaningful D-level × Quadratic interaction ($p = 0.50$) indicates that adding quadratic trends does not reveal differential learning trajectories between groups. Both high-D and low-D participants showed similar curvature; they differ only in their linear slopes. Given this, and the BIC's preference for the simpler model, we retained the linear specification. The key insight—that high-D participants show steeper declines in return percentages over time—is fully captured by the significant D-level × Round interaction in the linear model.

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